

BANK DETECTION SYSTEM

AN INTELLIGENT MACHINE LEARNING BASED APPROACH

SUBMITTED BY:
ANU SHREE KA

INTRODUCTION

- * Banking is an important part of the financial system and handles a large number of transactions every day.
- * With the growth of digital banking, identifying suspicious or fraudulent transactions has become increasingly important.
- * Machine Learning can analyze transaction data and identify patterns associated with potentially fraudulent activities.
- * This project aims to develop a **Bank Detection System** using Python and Machine Learning.
- * The system will classify transactions based on the patterns learned from historical data.

PROBLEM STATEMENT

- *The increasing use of digital banking has resulted in a large volume of financial transactions.
- *Detecting suspicious or fraudulent transactions manually is difficult and time-consuming.
- *Traditional rule-based methods may not identify complex transaction patterns effectively.
- *There is a need for an intelligent system that can analyze transaction data automatically.
- *Machine Learning can help detect potentially fraudulent transactions based on historical patterns.

OBJECTIVES

- *To develop a machine learning-based system for detecting potentially fraudulent bank transactions.
- *To collect and preprocess transaction data for analysis.
- *To identify important patterns and features associated with fraudulent transactions.
- *To train and evaluate a suitable machine learning classification model.
- *To classify transactions as **legitimate or potentially fraudulent**.
- *To evaluate the performance of the proposed system using suitable metrics.

Transaction Data → Preprocessing → ML Model → Prediction

EXISTING SYSTEM

- *Traditional banking systems mainly depend on predefined rules and manual monitoring.
- *Suspicious transactions are often identified using fixed thresholds or predefined conditions.
- *Manual analysis can be time-consuming when the number of transactions is very large.
- *Rule-based systems may have difficulty detecting new or complex fraud patterns.
- *Continuous monitoring and analysis can require significant human effort.

Limitations of Existing System

- ~Limited adaptability
- ~Time-consuming
- ~Difficult to handle large transaction volumes
- ~May miss previously unknown fraud patterns

PROPOSED SYSTEM

- *The proposed system uses **Machine Learning** to analyze bank transaction data.
- *Transaction data is first **preprocessed** to remove errors and prepare it for the model.
- *Relevant transaction features are used to train a **classification model**.
- *The trained model learns patterns from historical transaction data.
- *New transactions are classified as **Legitimate** or **Potentially Fraudulent**.
- *The system helps reduce manual effort and supports faster identification of suspicious transactions.

flow diagram

Transaction Data



Data Preprocessing



Feature Selection



ML Model



Fraud / Legitimate Prediction

SYSTEM METHODOLOGY

Banking Dataset



Data Preprocessing



Feature Selection



Machine Learning Model



Fraud/Legitimate Classification



Performance Evaluation

The proposed system processes banking transaction data, prepares the data for analysis, trains a machine learning model, and classifies transactions based on learned patterns. The model is then evaluated using suitable performance metrics.

DATASET & FEATURES

Dataset:

*The project uses a banking transaction dataset for detecting potentially fraudulent transactions.

*Each record represents a financial transaction.

*The dataset contains transaction details that can be used to identify patterns.

Important Features:

Transaction amount

Transaction type

Transaction time

Account/customer information

Transaction location

Previous transaction patterns

DATA PREPROCESSING

Data Cleaning: Remove duplicate and irrelevant records.

Missing Values: Identify and handle missing or incomplete data.

Data Transformation: Convert categorical information into a suitable numerical format.

Feature Scaling: Scale numerical features when required by the Machine Learning algorithm.

Data Splitting: Divide the dataset into **training and testing data**.

Data Validation: Check the prepared data before using it for model training.

Raw Dataset



Data Cleaning



Handle Missing Values



Data Transformation



Feature Scaling



Train/Test Split

Goal: Prepare clean and suitable data for accurate Machine Learning predictions.

MACHINE LEARNING MODEL

- *A suitable Machine Learning classification algorithm is selected for the project.
- *The model is trained using the preprocessed banking transaction data.
- *The trained model learns patterns from historical transactions.
- *It then predicts whether a transaction is **legitimate or potentially fraudulent**.
- *The model's performance is evaluated using standard classification metrics.

WORK FLOW

Preprocessed Data



Training Data



Machine Learning Classification Model



Prediction



Legitimate /  Potentially Fraudulent

MODEL TRAINING & TESTING

- *The preprocessed dataset is divided into **training and testing sets**.
- *The **training data** is used to teach the Machine Learning model.
- *The model learns patterns and relationships from previous banking transactions.
- *The **testing data** is used to check how well the trained model performs on unseen transactions.
- *The model's predictions are compared with the actual labels.
- *Performance is measured using suitable evaluation metrics.

SIMPLE DIAGRAM

Banking Dataset



Train-Test Split



Training Data



Testing Data



Train Model



Make Predictions



Performance Evaluation

Training → Learning patterns | Testing → Measuring performance

MODEL EVALUATION

- *The trained model is evaluated using the **testing dataset**.
- ***Accuracy** measures the overall percentage of correct predictions.
- ***Precision** measures how many predicted fraudulent transactions are actually fraudulent.
- ***Recall** measures how many actual fraudulent transactions are correctly identified.
- ***F1-Score** provides a balance between precision and recall.
- *A **confusion matrix** can be used to visualize correct and incorrect classifications.

Accuracy

Overall correctness

Precision

Correct fraud predictions

Recall

Fraud detection ability

F1-Score

Balance of precision & recall

RESULTS & ANALYSIS

- *The trained Machine Learning model is tested using unseen transaction data.
- *The model predicts whether transactions are **legitimate or potentially fraudulent**.
- *The performance of the model is analyzed using **Accuracy, Precision, Recall and F1-Score**.
- *A **confusion matrix** is used to understand correct and incorrect predictions.
- *The final results help determine the effectiveness of the proposed system.

MODEL RESULTS

Evaluation Metric	Result
Accuracy	To be calculated
Precision	To be calculated
Recall	To be calculated
F1-Score	To be calculated

Transaction Data → ML Model → Legitimate / Potentially Fraudulent

ADVANTAGES

Early Detection: Helps identify potentially fraudulent transactions at an early stage.

Automated Analysis: Reduces the need for manual checking of large numbers of transactions.

Fast Processing: Machine Learning can analyze transaction patterns efficiently.

Improved Accuracy: The model can learn from historical transaction patterns to improve predictions.

Scalable: The system can be adapted to handle larger transaction datasets.

Data-Driven Decisions: Provides useful insights based on transaction data.



Automated



Fast



Early Detection



Accurate



Scalable



Data-Driven

LIMITATIONS

Dataset Dependency: Model performance depends on the quality and size of the available dataset.

False Predictions: The system may sometimes classify a legitimate transaction as fraudulent or miss a fraudulent transaction.

Changing Fraud Patterns: New types of fraudulent behavior may not be recognized immediately.

Feature Dependency: The model's performance depends on the relevance and quality of the selected features.

Real-Time Challenges: A real-world banking system would require additional infrastructure for continuous real-time monitoring.

Security & Privacy: Banking data must be handled securely and protected from unauthorized access.

-  **Dataset Dependency**
-  **False Predictions**
-  **Changing Patterns**
-  **Feature Dependency**
-  **Real-Time Challenges**
-  **Security & Privacy**

FUTURE SCOPE

Real-Time Detection: Develop the system to analyze transactions in real time.

Advanced Machine Learning: Experiment with more advanced algorithms to improve detection performance.

Deep Learning: Explore neural-network-based approaches for complex transaction patterns.

Continuous Learning: Allow the model to learn from newly identified fraud patterns.

Real-Time Alerts: Generate immediate alerts when suspicious activity is detected.

Integration with Banking Systems: Connect the system with secure banking platforms for practical use.

simple upward-flow diagram:

Current System



Real-Time Detection



Advanced ML / Deep Learning



Continuous Learning



Banking System Integration

Goal: Build a smarter, faster and more adaptive fraud detection system.

Credit Card Fraud Detection Dataset

Why I'm choosing this one

It's much easier for us to understand than the huge 284,807-row dataset, while still giving us enough data to demonstrate:

Dataset → Preprocessing → Machine Learning → Prediction → Evaluation

The dataset includes features such as transaction amount, transaction hour, merchant category, location mismatch, device trust score, transaction velocity, and cardholder age

```
Bank_Detection_Project |  
├── dataset |  
├── credit_card_fraud_10k.csv |  
└── bank_detection.py
```

Data Preprocessing / Data Understanding

Dataset Analysis

Dataset contains 10,000 transaction records.

Each record represents a credit card transaction.

Dataset contains transaction-related attributes such as amount, transaction time and merchant category.

Missing values are checked before model training.

Duplicate records are identified and handled.

Data types are examined to prepare the dataset for machine learning.

IMPLEMENTATION:DATA LOADING & PREPROCESSING

```
import pandas as pd
```

```
# Load dataset
```

```
data = pd.read_csv("dataset/credit_card_fraud_10k.csv")
```

```
# Check missing values
```

```
print(data.isnull().sum())
```

```
# Check duplicate records
```

```
print(data.duplicated().sum())
```

```
# Display dataset information
```

```
print(data.info())
```

Under the code, write:

The dataset is loaded using Pandas. The data is checked for missing values, duplicate records, and data types before applying Machine Learning.

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier

# Convert categorical data into numbers
encoder = LabelEncoder()

for column in df.select_dtypes(include=["object"]).columns:
    df[column] = encoder.fit_transform(df[column])

# Separate features and target
X = df.drop("is_fraud", axis=1)
y = df["is_fraud"]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
model.fit(X_train, y_train)
```

IMPLEMENTATION: PREDICTION & EVALUATION

```
from sklearn.metrics import (  
    accuracy_score,  
    precision_score,  
    recall_score,  
    confusion_matrix  
)  
  
# Make predictions  
y_pred = model.predict(X_test)  
  
# Calculate performance  
accuracy = accuracy_score(y_test, y_pred)  
precision = precision_score(y_test, y_pred, zero_division=0)  
recall = recall_score(y_test, y_pred, zero_division=0)  
  
print("Accuracy:", accuracy * 100)  
print("Precision:", precision * 100)  
print("Recall:", recall * 100)  
  
# Confusion matrix  
cm = confusion_matrix(y_test, y_pred)  
print(cm)
```

Under the code: The trained model predicts the classes of unseen transactions. Accuracy, precision, recall, and the confusion matrix are used to evaluate the model.

MACHINE LEARNING DATA

Training records: 8000

Testing records: 2000

Number of features: 9

MODEL TRAINING COMPLETED

Random Forest model trained successfully.

PREDICTIONS COMPLETED

Predictions generated for testing data.

PREDICTIONS COMPLETED

Predictions generated for testing data.

MODEL EVALUATION

Accuracy : 99.8 %

Precision: 100.0 %

Recall : 86.67 %

CONFUSION MATRIX

[[1970 0]

[4 26]]

PS C:\Users\ANU\Desktop\BANK DETECTION>

CONCLUSION

The Bank/Credit Card Fraud Detection System demonstrates how Machine Learning can be used to identify potentially fraudulent transactions. The project uses transaction data, performs data preprocessing, and applies a Random Forest Classification algorithm to classify transactions as legitimate or fraudulent. The model is trained using historical transaction data and evaluated using Accuracy, Precision, Recall, and Confusion Matrix. The system provides an automated approach to fraud detection and can help reduce the effort required for manual transaction monitoring. Overall, the project shows that Machine Learning can be an effective tool for analyzing transaction patterns and supporting faster and more efficient fraud detection.